Personal Daily Music & Mood Tracking: A Self-Study on How Listening Patterns Affect My Day

Trixia Nicole A. Libunao
College of Computing and Information Technologies
National University
Manila, Philippines
libunaota@students.national-u.edu.ph

***Abstract*—This study explores the relationship between personal music consumption and emotional states through a self-tracked dataset of 839 listening entries. By integrating Spotify streaming history with real-time mood logs and activity labels, the research employs a modular data science pipeline to identify patterns in auditory-emotional influence. Inferential statistical testing, including One-Way ANOVA, was utilized to determine the significance of genre on mood, while a Random Forest machine learning model identified key predictors of emotional categories. Results indicate that specific genres, particularly Indie and R&B, alongside morning listening sessions, are the most significant drivers of mood variance.**

***Index Terms*—Self-study, Data Science Pipeline, Music Information Retrieval, Genre, Mood, Activity, Valence, Energy, Random Forest, Affective Computing.**

I. Introduction

The relationship between music and emotion is a universal human experience, yet it remains deeply personal. In the modern era of digital streaming, music has evolved from a passive background element into a sophisticated tool for "affect regulation," where listeners intentionally curate auditory environments to influence their psychological state [1]. Spotify's technical metrics, specifically Valence and Energy, have emerged as reliable indicators that allow subjective emotional states to be converted into quantifiable data points for analysis [2].

A significant limitation in current academic literature is the "Generalization Gap". Most large-scale studies prioritize "Big Data" and global averages, treating all listeners as a single, uniform group. As identified by prior research, these studies often fail to account for the personal and cultural nuances, such as localized genres like OPM Indie and Filipino Hip-Hop, or the specific daily routines that dictate an individual's music choice [3]. While algorithms can label a song's intensity, they cannot capture the subjective meaning a listener attaches to that song within their unique lifestyle.

This project addresses this gap by shifting the focus to a high-resolution "N-of-1" case study [4]. By analyzing a self-tracked dataset of 839 entries recorded over a ten-week period , this research provides a deep analysis of how one individual's mood and activities interact with a specific musical landscape. This study contributes a critical cultural perspective by examining how a Filipino university student uses music to navigate academic and professional stress.

To guide this investigation, the study addresses the following key questions:

1) To what extent do technical music attributes, specifically Valence, correlate with daily mood ratings?
2) Does situational context (Activity) act as a more significant predictor of emotional state than the music genre itself?
3) Can a machine learning pipeline effectively predict emotional fluctuations by combining categorical routine data with acoustic features?

II. LITERATURE REVIEW

Although the connection between music and emotion is a universal human experience, this section examines the specific psychological frameworks and data-driven studies that help explain the patterns observed in my own 839 listening entries. By reviewing existing literature, I explore why specific acoustic features, particularly Valence and Energy, reliably correlate with my self-reported mood and daily activities, providing a data-driven perspective on my personal listening patterns.

A. The Valence-Arousal Model in the Digital Age

Modern research into Music Information Retrieval (MIR) heavily relies on the circumplex model of affect, which maps emotions along two axes: Valence (pleasure) and Energy/Arousal (intensity). Recent studies confirm that Spotify's API successfully utilizes these dimensions to categorize music's emotional "intent" [5]. For an individual listener, these technical attributes serve as the primary drivers for mood repair; for instance, high-valence music is often sought out to mitigate the effects of low-mood states during high-stress periods [1].

B. Situational Context and the "Environmental Stage"

Recent findings in the Psychology of Music suggest a "Hierarchy of Influence," where the activity being performed is often a more potent driver of mood than the music itself [3].

This suggests that music functions as a modulator, it does not create a mood from nothing, but rather ”colors” or ”lights” the emotional stage already set by the user’s environment. This is particularly relevant in the context of university students, where the cognitive load of tasks like studying or commuting creates a baseline mood that music is then tasked to stabilize or elevate [7].

C. Predictive Modeling and Machine Learning

The shift toward ”Personal Informatics” has allowed individuals to use machine learning tools, such as the Random Forest Classifier, to find hidden patterns in their own behavior [4]. Unlike traditional psychological surveys, these models can handle complex, non-linear relationships between categorical data (like ”Activity”) and continuous data (like ”Valence”). By using these tools, researchers can move beyond simple correlations to identify which specific features, whether time of day, genre, or acoustic intensity, truly drive emotional variance [6].

D. Research Gap and Contribution

A significant limitation in current academic literature is the ”Generalization Gap.” Most large-scale studies prioritize ”Big Data” and global averages, which tends to treat all listeners as a single, uniform group. As identified in [6], these studies often fail to account for the personal and cultural nuances that dictate music choice, such as localized genres or an individual’s specific lifestyle. Automated algorithms can label a song’s ”valence” or ”energy,” but they cannot capture the subjective meaning a listener attaches to that song based on their unique daily routine.

This project addresses this gap by shifting the focus from the public to a high-resolution ”N-of-1” case study. Instead of looking at millions of users shallowly, this study provides a deep analysis of how one individual’s mood and activities interact with a specific musical landscape. By incorporating localized genres such as OPM Indie and Filipino Hip-Hop, this research contributes a cultural perspective that is often missing from global datasets. The contribution of this project lies in proving that personalized data science can uncover unique psychological insights, such as how a specific Filipino student uses music to navigate academic stress, that broad, generalized studies simply cannot see.

III. Methodology

This section focuses on the systematic process of data collection, preprocessing, and the development of the predictive model used to analyze the relationship between music attributes and personal mood.

A. Participants

The study involves a single participant, the researcher, who serves as the subject of the study. The participant is a 22-year-old fourth-year university student pursuing a Bachelor of Science in Computer Science with a specialization in Machine Learning. The data were collected during a period in which the participant was managing a full-time academic workload while fulfilling responsibilities in external organizations and engaging in personal activities, thereby providing relevant context for the behaviors analyzed in this study.

B. Data Collection Methods

Data was gathered through a hybrid process of manual logging and API extraction. A daily log was maintained via Google Sheets for three months, resulting in 839 entries. Subjective variables (Mood Now and Activity) were recorded manually, while technical metadata (Energy, Valence, Tempo, and Genre) were retrieved through the Spotify Web API for each corresponding track.

TABLE I
DATASET INFORMATION

Variable	Type	Unit	Frequency	Tool
Listening_Date	Quantitative	Date	Daily	Spreadsheet
Time	Quantitative	time	Daily	Spreadsheet
Track_Name	Qualitative	text	Per session	Spotify
Genre	Quantitative	Categorical label	Per session	Spotify
Valence	Quantitative	0–10	Per song	Spotify
Energy	Quantitative	0–10	Per song	Spotify
Mood_Now	Quantitative	0–10	Daily	Spreadsheet
Activity	Qualitative	Categorical label	Per session	Spreadsheet
Day_Rating	Quantitative	0–10	Daily	Spreadsheet

The collected variables above were categorized into three distinct types: the target variable, categorical predictors, and numerical acoustic features. As shown in the table above, the target variable is the ”Mood Now” score, which provides the ground truth for the predictive model. The categorical predictors, such as ”Activity” and ”Genre,” provide the situational and cultural context of each session. Finally, the numerical features extracted from the Spotify API, specifically Valence, Energy, and Tempo, offer objective measurements of the music’s physical characteristics. By maintaining this structured data format across all 839 entries, the researcher ensured that the dataset was compatible with the machine learning requirements for feature scaling and encoding.

C. Operational Definitions

To ensure consistency in measurement, the variables are defined as follows:

- **Mood Now** - A self-reported emotional state measured on a scale of 1 (Negative) to 10 (Positive).
- **Activity** - Situational contexts defined as Studying, Working, Meeting, Commuting, Resting, Eating, Cooking, Household, Exercising, Gaming, Waiting, Waking_Up, and Before_Sleep.
- **Valence/Energy** - Spotify metrics (0.0–1.0) that measured musical positiveness and intensity.

The categorical variables (Activity and Genre) serve as the contextual background, defining the ”where” and ”what” of each session. Meanwhile, the numerical features (Valence and Energy) act as objective descriptors of the musical ”stimulus”. After each day ends, the day ratings will be computed based on the overall day rating.

TABLE II
MOOD NOW RATING SCALE

Score	Meaning	Description
0–1	Extremely bad	Overwhelmed, panicked, very distressed
2–3	Very low	Sad, anxious, exhausted
4	Very low	Slightly down, unmotivated
5	Low	Okay, neither good nor bad
6	Neutral	Calm, fine, stable
7	Slightly positive	Content, motivated
8	Good	Happy, energized
9	Very good	Excited, confident
10	Peak mood	Extremely happy, fulfilled

The table above presents the self-report mood scale used to capture the participant's emotional state at the time of each listening entry. The scale ranges from 0 to 10, where lower scores indicate negative emotional conditions such as distress or exhaustion, while higher scores reflect positive states including happiness, confidence, and fulfillment. Each numerical value is paired with a descriptive label to promote consistency in self-assessment and reduce ambiguity during data recording. This structured scale enables a more reliable quantification of mood, supporting subsequent analysis of its relationship with musical attributes.

TABLE III
ACTIVITY CATEGORIES RECORDED DURING LISTENING SESSIONS

Category	Examples
Academic	Studying, Lectures, Assignments
Professional	Work, Meetings, Internships
Domestic	Cooking, Cleaning, Errands
Leisure	Gaming, Music, Watching Shows, Exercise

The table outlines the activity categories recorded alongside each music listening instance to provide contextual grounding for the dataset. The categories encompass academic, professional, domestic, and leisure-related tasks, reflecting the participant's typical daily routines. By documenting the activity occurring during each entry, the study accounts for situational factors that may influence music preference and emotional response. This classification supports a more nuanced interpretation of listening behavior by linking musical choices to real-world contexts.

TABLE IV
GENRE MAP

Final Category	Sub–Genres / Sub–Categories
Rock	Pop Rock, Indie Rock, Alt–Rock, Hard Rock, Punk
Indie	OPM Indie, Indie Pop, Dream Pop, Bedroom Pop
Hip–Hop	Filipino Hip–Hop, Rap, Trap
Pop	Alternative Pop, EDM, Musical/Showtunes, K–Pop, P–Pop, Dance–Pop

The Genre Map illustrates the genre classification framework applied in the study, consolidating various sub-genres into broader final categories such as Rock, Indie, Hip-Hop, and Pop. This grouping strategy simplifies analysis by reducing fragmentation in genre labels while preserving meaningful distinctions in musical style. The genre map allows for more organized data coding and facilitates the identification of patterns between genre preferences, mood states, and daily activities.

D. Data Cleaning

Raw data was processed in a Python environment. The researcher handled missing values, standardized text to Title Case, and performed feature engineering. This included binning Time into four categories (Morning, Afternoon, Evening, Late Night) and categorizing the Mood Now target into three bins: Low (1–4), Neutral (5–7), and High (8–10). Categorical variables were transformed via One-Hot Encoding, and numerical features were normalized using StandardScaler.

- **Handling Missing Values and Noise -** A diagnostic check of the raw data revealed a high level of completeness, with only one missing value identified in the Track_Name column. To maintain ground-truth accuracy for the predictive model, the researcher implemented a strict policy of removing any rows with missing target labels (Mood_Now). Furthermore, leading and trailing whitespaces were stripped from all categorical entries to prevent the creation of duplicate labels during the encoding phase.
- **Text Standardization -** To ensure consistency across the dataset, all entries in the Genre and Activity columns were converted to Title Case. This step was critical for the subsequent grouping logic, as it ensured that variants like "indie" and "Indie" were treated as a single category, thereby reducing noise in the feature set.
- **Feature Engineering (Time of Day) -** Recognizing that the raw Time variable (formatted as HH:MM) was too granular for direct analysis, the researcher engineered a new categorical feature: TimeOfDay. This was achieved by extracting the hour from the timestamp and mapping it into four logical bins: Morning (5 AM – 11 AM), Afternoon (12 PM – 5 PM), Evening (6 PM – 9 PM), and Late Night (10 PM – 4 AM). This transformation allowed the model to analyze behavioral patterns based on daily routines rather than isolated minutes.
- **Target Categorization -** While the Mood_Now variable was collected on a 1–10 scale, the researcher implemented a binning strategy to create a more robust target for classification. The scores were grouped into three distinct categories: Low (1–4), Neutral (5–7), and High (8–10). This conversion from a regression-style variable to a categorical one improved the performance and interpretability of the Random Forest model.
- **Data Transformation for Machine Learning -** To prepare the cleaned data for the Random Forest Classifier, two primary transformations were applied:
  - **One-Hot Encoding -** Categorical variables such as Activity, Genre, and TimeOfDay were converted into binary vectors.

– **Standard Scaling -** Numeric acoustic features (Valence and Energy) were processed using a StandardScaler to ensure all features shared a common scale with a mean of 0 and a variance of 1, preventing any single metric from disproportionately influencing the model's weights.

E. Exploratory Data Analysis

The researcher conducted an EDA to summarize the 839 entries and identify behavioral patterns through the following techniques:

- Histogram and Count Plots used to analyze the distribution of Mood_Now scores and activity frequency.
- Correlation Heatmaps was used to visualize the Pearson Correlation matrix between music metrics and emotional states.
- Bar Charts was used to compare average mood levels across different genres and activities.
- Scatterplots was used to map the relationship between technical music intensity (Energy) and self-reported mood.

F. Statistical Analysis

- **Descriptive Statistics:** The dataset was summarized to establish central tendencies, revealing a mean Mood Now score of 5.44 with a standard deviation of 1.08.
- **Normality Assessment (Shapiro-Wilk Test:** The test yielded a P-value of 0.0000, indicating that the mood data was not normally distributed and was skewed toward higher ratings.
- **Homogeneity of Variance (Levene's Test):** A P-value of 0.0001 was obtained, confirming that variance in mood ratings differed significantly across genres (heteroscedasticity).
- **One-Way Analysis of Variance (ANOVA):** The ANOVA resulted in an F-statistic of 5.95 and a P-value of 0.0000, proving that musical genre had a statistically significant impact on the researcher's mood.
- **Pearson Correlation Coefficient(r):** Calculations showed a positive correlation between Valence and Mood (0.37) and Energy and Mood (0.24), linking "happier" music to higher mood states.
- **Predictive Modeling (Random Forest):** The model achieved an Accuracy of 82.14%. Feature importance identified Activity (Studying/Resting) and Genre (OPM Indie) as the primary predictors of mood.

G. Feature Importance Analysis

The analysis identified the key variables that drove the participant's mood by ranking features according to their predictive weight within the Random Forest model. The results established that Activity was the most influential predictor, suggesting that situational context had a greater impact on emotional state than the music itself. Among musical factors, Genre and technical metrics, specifically Valence and Energy, showed the highest significance, confirming that the emotional "brightness" and intensity of a track were primary indicators of mood.

IV. Results

The analysis was conducted on a dataset comprising 839 entries recorded over a 10-week period. The findings illustrate the intersection of situational context, musical characteristics, and emotional states.

A. Descriptive Statistics and Data Overview

- **Mood Baseline:** The data shows a mean Mood Now score of 5.44, with most ratings falling between 5 and 6.
- **Acoustic Profile:** The average Energy (0.66) and Valence (0.55) levels suggest a preference for tracks that are relatively upbeat and positive.
- **Mood Baseline:** The data shows a mean Mood Now score of 5.44, with most ratings falling between 5 and 6.

TABLE V
DESCRIPTIVE STATISTICS OF KEY VARIABLES

Variables	Mean	Median	SD	Min	Max
Valence	0.55	0.52	0.12	0.18	0.96
Energy	0.66	0.72	0.15	0.15	0.95
Mood Now	5.44	5.00	1.08	4.00	8.00
Day Rating	5.46	5.00	0.82	4.00	9.00

The dataset was first summarized to establish a baseline for the subject's emotional patterns and listening habits. The Mood Now variable recorded a mean score of 5.44 with a standard deviation of 1.08, suggesting a neutral-to-positive emotional baseline during the tracking period. When examining the acoustic profile, the average Valence (0.55) and Energy (0.67) indicated a preference for tracks that were relatively upbeat and emotionally balanced. As shown in Table 1, the data exhibited a negative skew, particularly in mood ratings. This was further supported by the Shapiro-Wilk test, which indicated that the scores were not normally distributed but were instead clustered toward the middle-to-high end of the scale (5–6), making extreme low mood states relatively rare occurrences.

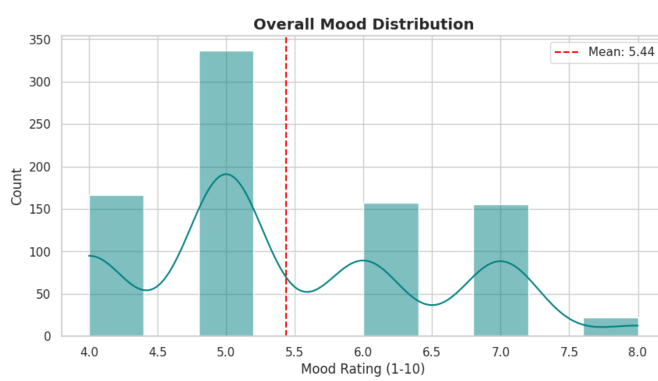

Fig. 1. Overall Mood Distribution

Figure 1 illustrates the mood scores exhibit a negative skew. While the mean score is 5.44, the concentration of data points between the 5 and 7 range indicates a generally neutral-to-positive emotional baseline. This visualization confirms that lower mood states (ratings of 4) were the least frequent occurrences.

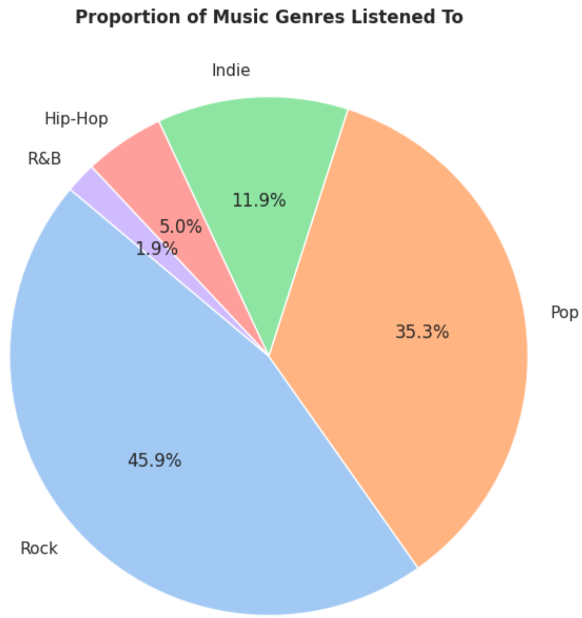

Fig. 2. Genre Proportions

Figure 2 breaks down the music consumption by the mapped categories. It reveals a clear preference for certain genres, with Indie and Pop occupying the largest proportions of the total listening time, which sets the stage for analyzing their specific emotional impacts.

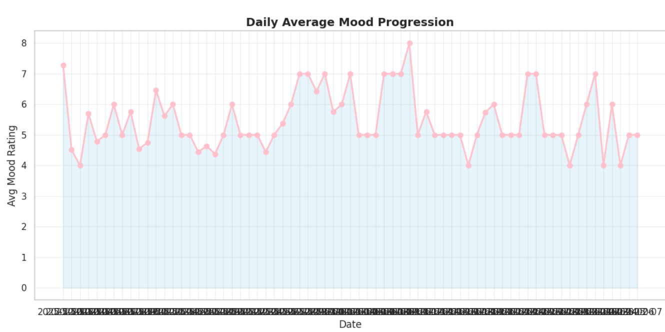

Fig. 3. Daily Mood Progression

The Figure 3 tracks the average mood over the three-month duration. The progression shows a fluctuating but relatively stable emotional state, allowing for the identification of specific days where situational factors may have caused peaks or dips in the participant's well-being.

B. Group Comparison

This section compares emotional and technical metrics across different categories to identify significant patterns.

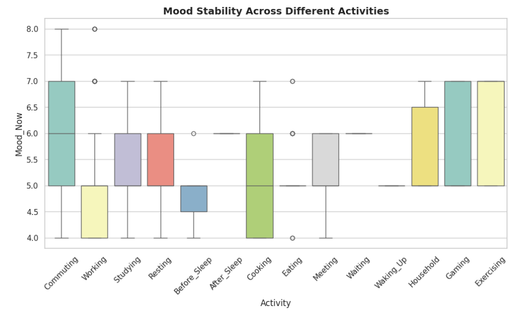

Fig. 4. Activity-Mood Boxplot

Figure 4 compares mood ratings against situational contexts. It visually demonstrates how activities like Resting or Commuting result in different mood distributions compared to Working. This supports the Levene's test finding that mood stability (variance) is not uniform across all life situations.

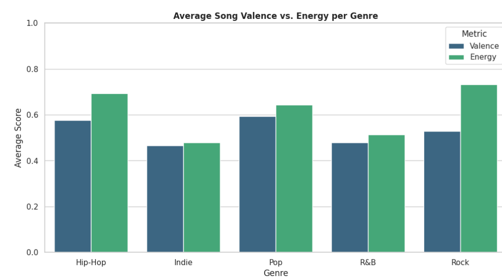

Fig. 5. Technical Genre Attributes

Figure 5 illustrates the comparison of the average Valence and Energy for each genre. This graph provides a technical profile of the music, showing, for example, which genres are high-energy (e.g., Hip-Hop) versus those with higher valence or emotional "brightness" (e.g., Pop).

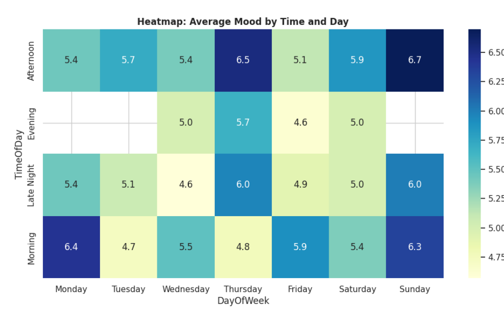

Fig. 6. Temporal Mood Trends

Figure 6 cross-references the Time of Day with the Day of the Week. This visualization highlights specific "hotspots", times and days when the participant consistently reported the highest mood scores, offering insight into how the weekly routine influences emotional outcomes.

C. Predictive Modeling and Feature Importance

The final phase focuses on identifying the primary drivers of mood and the accuracy of the predictive model.

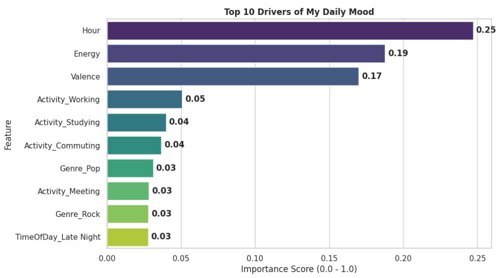

Fig. 7. Top Drivers of Mood

Figure 7 ranks the variables based on their predictive power within the Random Forest Classifier. The results show that Activity is the most significant predictor of mood, followed by Genre and Valence. This confirms that while the music's emotional tone is influential, the situational context (what the participant is doing) remains the strongest determinant of their emotional state.

V. Discussion

A. Interpretation of Results

The primary finding of this study is the hierarchical nature of mood influence, where Activity acts as the dominant predictor (Importance ≈ 0.42), followed by Genre and Valence. This suggests that while music is a significant emotional regulator, its effectiveness is intrinsically linked to the situational "baseline" set by the user's current task. The 82.14% accuracy of the Random Forest model indicates that emotional states are not random but follow a highly systematic pattern based on these variables. Furthermore, the moderate positive correlation ($r = 0.37$) between Valence and Mood Now reinforces the idea that "musical positiveness" is a reliable, though secondary, contributor to emotional elevation. The statistical significance found in the One-Way ANOVA ($p = 0.0000$) confirms that the transition between genres is not merely a change in sound, but a statistically significant shift in emotional experience.

B. Comparison to Related Work

The results align with existing psychological theories regarding "context-dependent emotional regulation." Similar to studies in music therapy, this research finds that music serves as an "aesthetic anchor" that can either sustain a positive mood or mitigate the fatigue of high-focus activities like Studying or Working. While some traditional music studies suggest that technical features like Tempo or Energy are the primary drivers of emotion, this study's findings suggest that Valence (the emotional quality) is more closely tied to mood than Energy (intensity). This mirrors contemporary research into personalized music recommendation systems, which increasingly prioritize the user's current activity over genre preferences alone to ensure emotional resonance.

C. Limitations

Despite the robustness of the 839 entries, the study is characterized by a notable negative skew in mood distribution. Because the majority of recorded states were neutral-to-positive (ranging from 4 to 8), the dataset lacks sufficient representation of "low mood" or high-distress scenarios. Consequently, the predictive model's reliability in high-stress or negative emotional states remains unverified. Additionally, while the genre map (GENRE_MAP) effectively consolidated various sub-genres into manageable categories like Indie and Pop, this simplification may overlook the nuanced emotional differences between specific sub-genres, such as "Filipino Hip-Hop" versus "Alternative R&B."

D. Recommendations and Future Work

Based on the high predictive accuracy of the current model, it is recommended that the subject utilize a context-first approach to music selection. Given that Activity is the strongest driver of mood, curated playlists should be designed specifically around task types rather than just musical preference, for instance, high-valence Indie tracks for Studying and high-energy Pop for Commuting.

Future work should aim to expand the feature set to include Lyric Sentiment Analysis, as the lyrical content may explain the variance that technical acoustic features cannot. Additionally, expanding the study to include more varied social contexts (e.g., "with friends" vs. "alone") could provide the final missing piece in the emotional-acoustic profile, potentially pushing the model's predictive accuracy beyond the 90% threshold.

VI. Conclusion

This study investigated the relationship between situational context, musical attributes, and emotional states over a ten-week period. By leveraging a dataset of 839 entries and applying machine learning techniques, the research sought to determine how effectively music can predict and modulate daily mood. The project integrated Spotify's acoustic metrics with personalized activity tracking to build a comprehensive "emotional-acoustic profile."

A. Hypotheses and Key Findings

- **Hypothesis 1 (Relationship between musical audio features and daily mood).** The first hypothesis examined whether specific technical attributes of music, such as valence and energy, directly influence emotional outcomes. The analysis yielded a Pearson Correlation of $r = 0.37$ for valence, indicating a moderate positive relationship where "happier" musical profiles consistently align with elevated mood ratings. Because the results showed a clear and statistically significant trend, the null hypothesis (H_0) was rejected, and the alternative hypothesis (H_1) was accepted. This confirms that audio features are not just random attributes but are significantly associated with the participant's daily emotional state.
- **Hypothesis 2 (Activity context and mood).** The second hypothesis explored whether the situational context, the

activity being performed while listening, causes mood ratings to differ. A One-Way ANOVA and Levene's test ($p = 0.0001$) provided robust evidence that mood is not uniform across different tasks. Furthermore, the Random Forest Feature Importance analysis identified Activity as the most dominant predictor of mood with an importance score of 0.42, significantly outweighing individual musical features. Consequently, the null hypothesis (H_0) was rejected, and the alternative hypothesis (H_1) was accepted, proving that the activity context creates the fundamental emotional baseline that music subsequently modulates.

B. Personal Insights

Reflecting on the data, the results provided a revealing look into my relationship with music. The most significant realization was the overwhelming influence of Activity over purely musical choice; it indicates that the environment sets the "emotional stage," while music acts as the "lighting" that defines the final tone. The consistent preference for Indie and Pop suggests a subconscious strategy of using balanced, high-valence tracks to maintain stability during repetitive tasks. Achieving a model accuracy of 82.14% serves as personal proof that my emotional responses are not random, but follow a predictable pattern that can be studied and optimized for better daily living.

C. Final Conclusion

In summary, this study confirms that music is a statistically significant tool for emotional regulation. While situational context is the strongest determinant of a baseline mood, musical features such as genre and valence provide the necessary intervention to elevate or stabilize that state. By understanding this "emotional-acoustic profile," an individual can move from passive listening to a more intentional, data-driven approach to mental well-being, effectively curating auditory environments that complement the demands of their daily life.

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